Inspired by recent works such as [30, 12], we generate the training data using a music playlist dataset and an LLM. These synthetic datasets have been shown to closely resemble human conversations in qualitative evaluations and have proven effective for training recommendation models. Following our primary reference [12], each conversation is generated as a journey of machine-assisted music discovery; assistants continue to react by choosing music items that satisfy the user's request. A notable difference is that in TALKPLAY, the base dataset is not simply a music dataset (e.g., [4]) but a playlist dataset, in order to simulate more plausible and coherent conversations and recommendations.

3.1 Base Dataset: Million Playlist Dataset

In the current era of music streaming, playlists are playing a crucial role in the music listening experience, providing unique and direct information about music items: which music items go along together well, according to the real listeners. They may suit well for the purpose of training TALKPLAY as well, as a base dataset, since conversation consisting of tracks in the same playlist would be somewhat diverse yet coherent, and hence realistic.

Table 1: Statistics of the TALKPLAY dataset.

Dataset	Training	Evaluation
Million Playlist Dataset (MPD)		
# of Playlists	999,000	1,000
# of Average Tracklist	63.73	59.41
# of Warm Tracks	1,714,772	37,114
# of Cold Tracks	-	2,357
# of Vocab	24,129	713
TalkPlay Dataset		
# of Playlist	116,099	1,000
# of Dialogue	531,552	1,000
# of Tracks	405,543	11,006
# of Vocab	317,705	19,736
Avg. # of turns	6.95	13.05

The Million Playlist Dataset (MPD) is chosen as our base dataset [8]. Since being introduced in 2018, this dataset remains one of the largest playlist datasets available to the public by including one million Spotify playlists. The MPD provides track metadata and playlist co-occurrence and is extended with audio content, lyrics, and semantic annotations. The audio content are successfully crawled through the Spotify API for 1,717,148 items. The lyrics are extracted using WHISPER-LARGE-V3 [38], a well-known speech transcription model that performs decent lyrics transcription. For semantic annotations, we leverage pretrained music understanding models to extract comprehensive musical attributes. Specifically, we use LP-MUSICCAPS [13] to extract high-level semantic features including genres, moods, instruments, and vocal styles. Additionally, we employ MADMOM [5] to extract structural music properties such as key signatures and tempo information. This multi-faceted approach ensures our model has access to a rich representation of musical characteristics.

3.2 TalkPlay Dataset

We use an LLM and the base dataset (MPD) to generate TalkPlay Dataset, a music conversation dataset with natural language and music recommendation.[3] These are the key design principles and choices to ensure the synthetic data can be used for the training of TALKPLAY.

Coherence: Given a user query/reaction, system should recommend music with relevant dialogue.

Modality Coverage: Conversations span four target modalities: audio, lyrics, metadata, and semantic tags. The playlist co-occurrence is excluded because: i) this modality is implicit and difficult to verbalize, and ii) in our task setup, playlist co-occurrence would be inherently captured (Section 2.1).

User Simulation: The user may skip or reject the recommended tracks, requesting for alternative recommendations. For rejected tracks, we employ hard negative sampling by using songs from artists whose tracks appear in the playlist. This approach ensures that the rejected recommendations are plausible but not suitable for the current conversation context.

Text Format: The LLM output, the synthesized conversation, should be in a structured format. The JSON format is chosen and used as presented in Section B.2

Based on this, GEMINI-1.5-FLASH-002 [44] is used to create music conversation given a prompt that includes i) our goal and the aforementioned conditions, ii) track identifiers and their metadata, semantic tags, lyrics. The prompt is included in Section B.1. Table 1 compares our dataset with the base dataset. Using 116k playlists with an average of 7 tracks each, our synthesis approach yields 532k multi-turn music recommendation dialogues. By leveraging rich textual annotations

[3] We will release the dataset to the public after the review period.

of music sequences, our LLM-based data synthesis technique significantly expands the text-music associations beyond just playlist titles, increasing the token count from 24k to 318k. Each user-assistant interaction in our synthetic conversation dataset follows a triplet structure: {USER QUERY, RECOMMENDED MUSIC, ASSISTANT RESPONSE}. The user query and assistant response are natural language expressions, while the recommended music is represented by a unique track identifier.

Data Split: To reflect the real-world recommendation scenarios, we perform a chronological data split based on the playlist creation date. The test set is created by randomly sampling 1,000 playlists from the last creation date (2017-11-01), and training set include all the rest of the tracks. Through this split, around 2,357 tracks in the test split are cold-start items, tracks that only appear in the test set. Assuming a real-world scenario, these items represent newly released songs, which traditional collaborative filtering algorithms have no information to make recommendation from. In contrast, being multimodal, TALKPLAY is able to handle these items by understanding content such as audio, lyrics, metadata and semantic features.

4 Experiments

We conduct a comprehensive evaluation of TALKPLAY, focusing on two critical dimensions: (1) retrieval performance - the model's ability to recommend relevant music items based on multi-turn dialogue context, and (2) response quality - the model's capability to generate contextually appropriate, natural language responses that explain recommendations. Through both objective metrics and human evaluation, we examine how our generative approach compares to traditional retrieval methods across various conversation scenarios and user preferences.

Objective Evaluation: For the multi-turn recommendation evaluation, we simulate a turn-taking scenario where each turn is conditioned on the previous turns and the current turn's user query, with the goal of retrieving appropriate music recommendations. Following previous works [7, 12], we employ standard ranking metrics - Mean Reciprocal Rank (MRR) and Hit Rate at K (Hit@K). We evaluate retrieval performance for each turn using 1,000 multi-turn conversations derived from 1,000 playlists in Table 1.

Subjective Evaluation: To assess the real-world usability of TALKPLAY, we conducted rigorous A-vs-B human preference evaluations against baseline models and ground truth. In each evaluation scenario, evaluators compared a pair of outputs (recommendations and responses) from different models based on a shared conversation context, to choose a preferred model (or a tie) on the two critical dimensions: (1) *Recommendation Relevance* - how well the recommended music items aligned with the conversation context, and (2) *Response Naturalness* - the conversational quality and contextual appropriateness of the generated text responses. We recruited 18 evaluators with expertise in music research or professional experience in the music industry. Each evaluator assessed 10 randomly selected conversation scenarios, yielding 180 total ratings.

Training Details: Using the synthetic conversation dataset (Section 3), TALKPLAY, a vocabulary-expanded Llama-3.2-1B model, is trained with a 1024-token context length, a learning rate of $1e-4$, and the AdamW optimizer [32]. The model reported in this paper is trained on 8 GPUs with a total batch size of 48 items (or 49,152 tokens).

Baseline Models: We evaluate our model against three baseline approaches: BM25 [40] for classical text-based sparse retrieval, NV-Embeds-V2 [29] for deep similarity retrieval with LLM embeddings, and CLAP [49] for multimodal audio-text retrieval. We also include two generative retrieval models, SASRec [26] and TIGER [39]. While both models perform next token prediction for recommendation, SASRec represents each of the 400k tracks as unique IDs, while TIGER uses Semantic IDs that combine SentenceT5 [35] embeddings with residual vector quantization (RVQ) for track representation. Unlike TALKPLAY, SASRec and TIGER lack natural language understanding capabilities. As a result, these models can only make recommendations based on track sequences, ignoring the semantic content of user queries in multi-turn conversations. Additional details can be found in Appendix D.

For our subjective evaluation, we chose BM25 [40] and NV-Embeds [29] as baseline models, as they demonstrated strong query understanding capabilities in our objective evaluation (In Table 2). To enable these retrieval-based models to generate conversational responses, we implemented a two-stage CRS using Llama-3.2 1B [16]. This pipeline first retrieves relevant music items using retrieval model, then uses these retrieved items as context for the LLM to generate appropriate responses.

Table 2: Comparison of recommendation performance across different models. MRR indicates the mean reciprocal rank and Hit@K measures proportion of relevant items in top-K results. QU and RG indicate query understanding and response generation capabilities, respectively.

Models	Retrieval Methods	QU	RG	MRR(↑)	Hit@1(↑)	Hit@100(↑)
Zero-shot Recommendation Models						
BM25 [40]	Sparse	✓	✗	0.030	0.002	0.265
CLAP-Music [49]	Similarity-based	✓	✗	0.011	0.003	0.153
NV-Embed-V2 [29]	Similarity-based	✓	✗	0.029	0.005	0.255
Finetuned Recommendation Models						
CLAP-Music [49]	Similarity-based	✓	✗	0.017	0.005	0.189
NV-Embed-V2 [29]	Similarity-based	✓	✗	0.032	0.005	0.281
SASRec [26]	Generative	✗	✗	0.002	0.000	0.008
TIGER [39]	Generative	✗	✗	0.008	0.001	0.092
TALKPLAY (Ours)	Generative	✓	✓	**0.049**	**0.026**	**0.288**

5 Results

5.1 Conversational Music Recommendation Results

Our results in Table 2 demonstrate that TALKPLAY significantly outperforms both zero-shot and finetuned baseline models across all key metrics. TALKPLAY achieves an MRR of 0.049, which is higher than the best baseline (finetuned NV-Embed-V2 at 0.032). More notably, TALKPLAY shows a dramatic improvement in Hit@1 performance (0.026), achieving more than 5 times the accuracy of the next best model. This demonstrates a strong correlation between the cross-entropy loss function used in next token prediction and the Hit@1 metric. The superior Hit@1 performance of TALKPLAY suggests that our generative approach can directly produce highly relevant recommendations without requiring extensive candidate consideration in the realistic music listening experience.

Among zero-shot recommendation models (upper side of Table 2), traditional sparse retrieval methods such as BM25 and embedding-based approaches like NV-Embed-V2 exhibit comparable performance, with MRR scores of 0.030 and 0.029 respectively. This indicates that, in the absence of task-specific training, dense embedding method [29] offer limited benefit over simpler retrieval techniques [40]. One reason for this close performance is that a large portion of music-related queries involve metadata—such as track titles, artist names, and album information—where sparse retrieval methods like BM25 are particularly effective due to their strength in exact token matching. In contrast, CLAP [49] performs significantly worse in multi-turn recommendation scenarios (MRR of 0.011). Given that CLAP was trained primarily on semantic annotations like genre and instruments without conversational context, this suggests its limited dialogue understanding capability leads to lower performance in conversation-based recommendation scenarios.

In fine-tune recommendation models (lower side of Table 2), fine-tuning improves performance for all baseline models, but the gains are modest compared to TALKPLAY's results. Notably, the sequencial recommendation models, such as SASRec [26] and TIGER [39], perform significantly worse than retrieval-based approaches. It indicates that the challenge of generating accurate recommendations without query understanding capabilities in conversational recommendation scenario.

Figure 2 shows how retrieval performance varies across conversation turns. The MRR scores reveal an interesting pattern: while TALKPLAY starts with comparatively lower scores in the first 1-2 turns, it significantly outperforms all baselines from Turn 4 onward and maintains relatively stable performance even as the conversation extends to later turns to Turn 13. In contrast, baseline methods exhibit notable degradation as the conversation continues. This pattern shows a fundamental trade-off in conversational recommendation

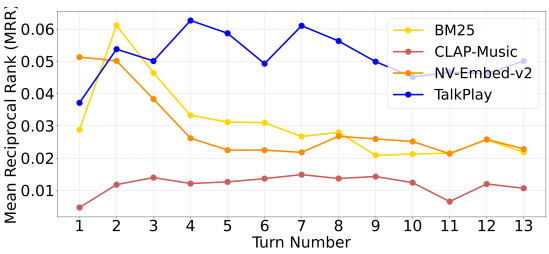

Figure 2: Performance comparison across conversation turns. The x-axis shows the turn number in the dialogue, and the y-axis shows Mean Reciprocal Rank (MRR).

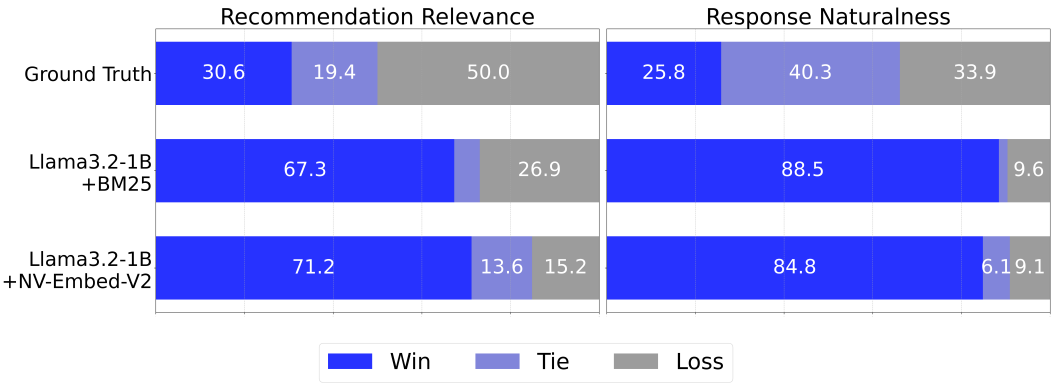

Figure 3: A-vs-B human evaluation results, comparing TALKPLAY against existing conversational music recommendation models on recommendation relevance (left) and response naturalness (right).

systems: TALKPLAY requires a few dialogue exchanges to establish user preferences and context, but once established, it excels at maintaining this understanding throughout extended interactions.

This result also emphasizes the fundamental challenge of multi-turn recommendation. Similarity-based retrieval methods must rely on simple aggregation strategies (e.g., averaging) to combine the previous chat history, which becomes less effective as context grows more complex. In contrast, TALKPLAY's LLM-based approach leverages the model's internal attention mechanism during token prediction to naturally capture and prioritize relevant contextual signals throughout the conversation. This enables more nuanced understanding of evolving user preferences, particularly beneficial in real-world music discovery scenarios where conversations typically extend beyond a few initial exchanges.

5.2 Qualitative Results

Figure 3 presents the results of our A-vs-B human evaluation, which compares TALKPLAY against both ground truth data and a set of baseline models. We conducted a human evaluation study to assess the quality of our model's recommendations and responses. Participants were presented with multi-turn conversation scenarios and asked to compare recommendations and generated responses between two models. The survey interface is showed in the appendix C.

In the comparison with ground truth data,TALKPLAY demonstrates competitive performance: it achieves a combined *Win+Tie* rate of 50% for the *Recommendation Relevance* metric and 66.1% for *Response Naturalness*. These scores indicate that the outputs generated by TALKPLAY are on par with ground truth recommendations and responses in terms of both relevance and fluency.

When compared against baseline models, TALKPLAY exhibits a clear performance advantage. Specifically, in the domain of recommendation relevance, our model outperforms both baselines: Llama3.2-1B + NV-Embeds-V2 with a Win Rate of 71.2% and Llama3.2-1B + BM25 with a Win Rate of 67.3%. These results validate the superiority of our LLM-based approach over traditional retrieval-based baselines in multi-turn recommendation scenario.

The improvements are even more substantial in the response generation task, underscoring the proposed model's capability to generate more natural and contextually appropriate replies within music-related conversations. Human evaluators consistently preferred our model's responses, with Win Rates of 84.8% and 88.5% over those generated by the general-purpose Llama-3.2-1B Instruct model.

5.3 Multimodal Feature Importance

Tables 3 and 4 present comprehensive ablation studies that systematically analyze the relative importance of different modalities and their weighting strategies in our model. Our analysis of weighting strategies in Table 3 reveals a clear pattern: approaches that emphasize coarse-grained features (playlist and semantic tokens) over fine-grained features (lyrics and audio tokens) consistently demonstrate superior performance. Specifically, both linear and quadratic coarse-to-fine weighting schemes achieve an MRR of 0.049, representing a substantial 29% improvement over the uniform